

SAFETY HAZARDS IN CONSTRUCTION

Walking and Working Surfaces, including Fall Protection



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U.S Bureau of Labor Statistics

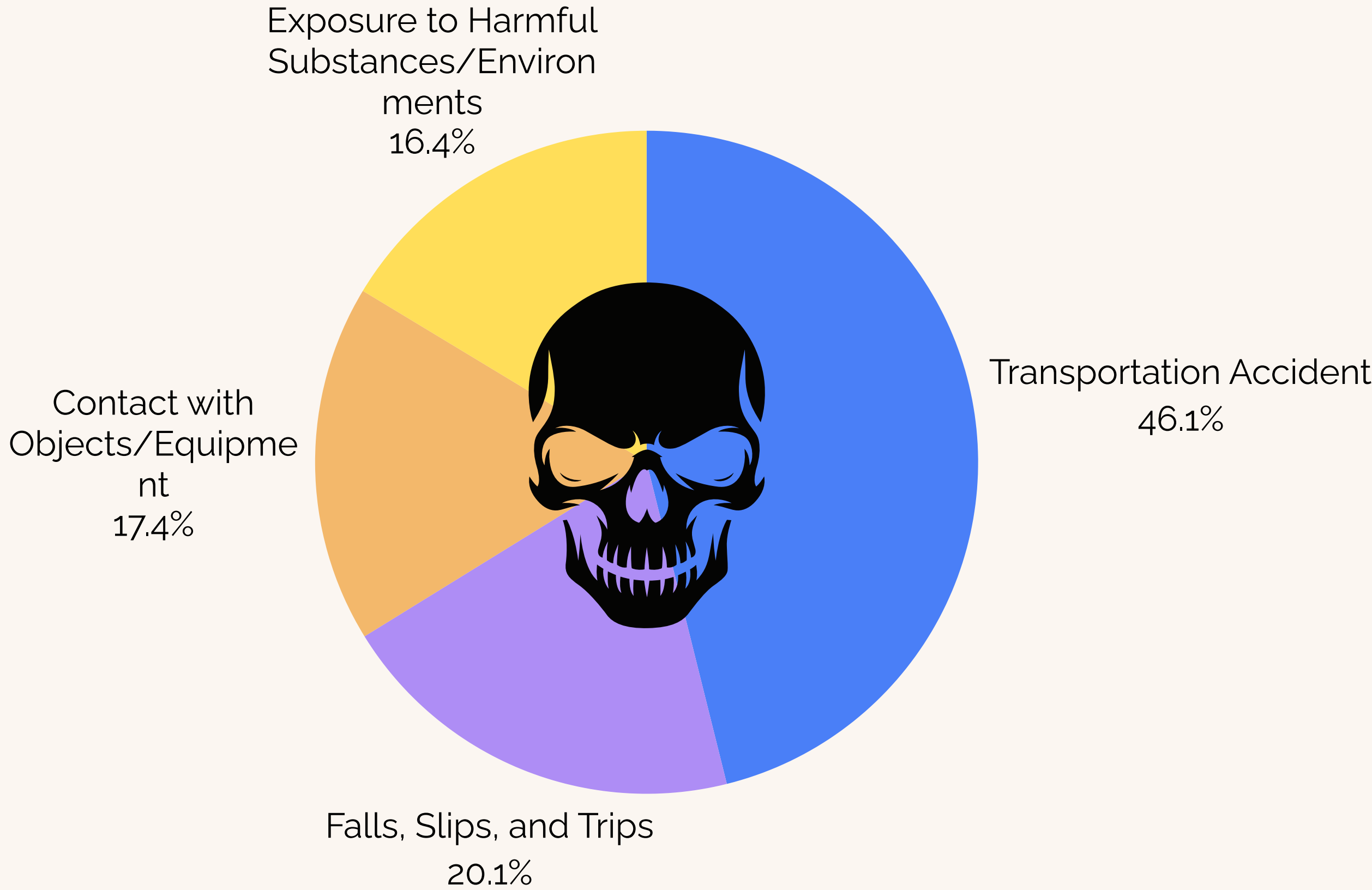
Accidental Deaths



Accidental Deaths
4,957

U.S Bureau of Labor Statistics

Accidental Deaths



Lesson objectives:

- 1. Identify hazards in the workplace associated with walking and working surfaces.**
- 2. Identify best practices for eliminating or controlling hazards associated with walking and working surfaces in the workplace.**
- 3. Recognize Employers must provide documented fall hazard to protect workers from walking and working surface hazards.**



Slips & Trips Also Falls



01 Slips

Slip primarily means to lose your balance, slide unintentionally, or make a minor mistake.

02 Trips

A foot catches on an object or uneven surface, disrupting your balance and usually causing you to stumble or fall forward.

03 Falls

A fall accident is defined as any unintentional event that causes a person to lose balance, slip, or trip, resulting in them coming to rest inadvertently on the ground, floor, or another lower level.

SLIP HAZARD



01 Trip Hazards

- Poor Housekeeping
- Loose Flooring
- Uneven Surfaces
- Potruding Items

02 How to Control them?

- Workplaces must be well lit, clean, and marked
- Materials and scraps must be kept in organized storage
- Trip hazards must be repaired and/or reported
- Hoses or long cords must be rerouted to make sure to keep clear from walkways.

TRIP HAZARD



01 Trip Hazards

- Grease
- Oil
- Water
- Ice
- Snow
- Liquids
- Spills
- Polished Floors

02 How to Control them?

- Keep walking/working surfaces as clean and dry as possible.
- Footware must be slip resistant
- Operations that requires water must have a appropriate drianage system
- Clean up or mark and report spills.
- Remove ice and snow frequently and regularly

FALL HAZARD



01 Fall Hazards

- **Elevated Surfaces**
 - Towers
 - Tall Machineries
 - Platforms
- **Low Level Surfaces**
 - Open pits
 - Open tanks
 - Vats
 - Ditches
- **Structural Collapse**
 - Structurally unsound surfaces
 - Structures with exceeding load limits

FALL HAZARD



02 How to Control them?

- For Elevated Surfaces
 - Temporarily or permanent always use guardrails
 - If possible use PFAS
 - Personal Fall Arrest System
 - Keep items away from or near the ledge to avoid falling hazards
- Platforms
 - Use guardrails system at all times
 - If a platform is near machinery must have railings regardless of height
- Structural Collapse
 - Ensure the surface is structurally sound
 - The surface must be able to support potential load which includes people, equipment and materials
 - Load limits must be known or posted

FALL HAZARD



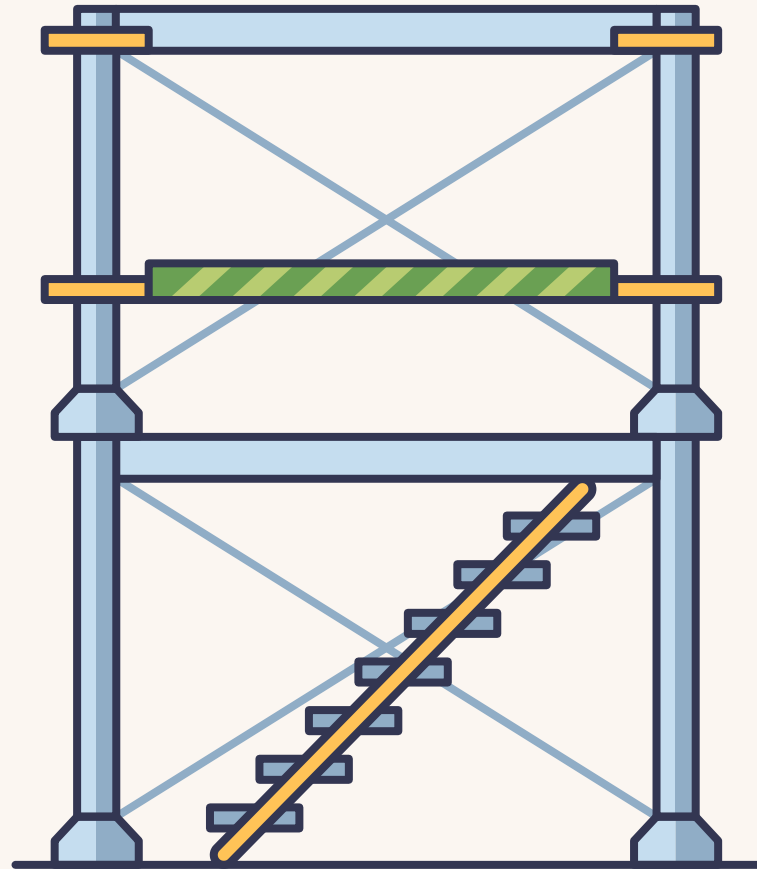
02 How to Control them?

- Personal Fall Arrest System
 - Know the ABCs
 - Anchorage
 - Body Harness
 - Components

Conditions that may lead to a Fall



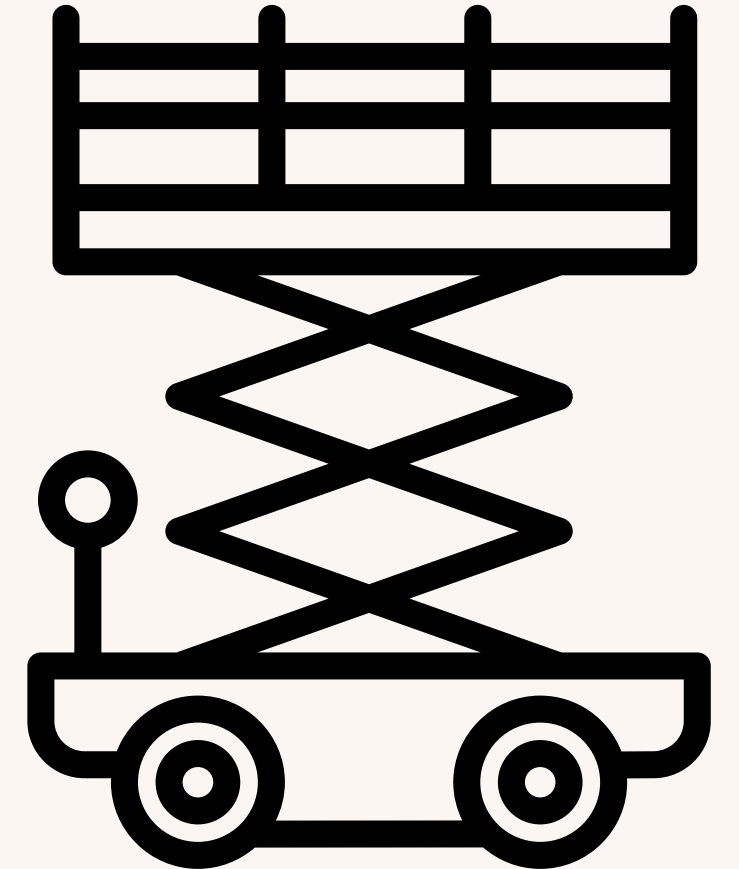
LADDERS



SCAFFOLDS



STAIRWAY



SCISSORLIFT

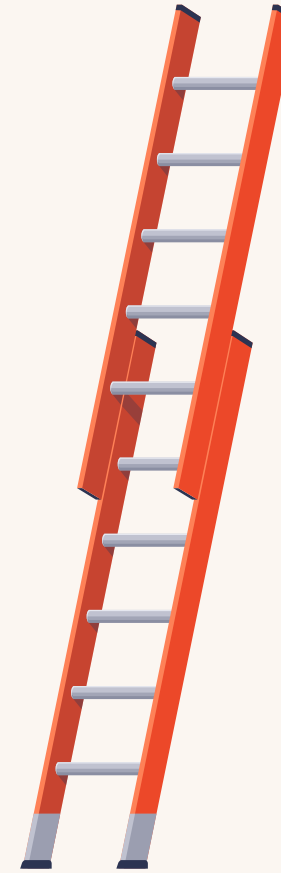
Conditions that may lead to a Fall



LADDERS



Step Ladder



Extension Ladder



Fized Ladders

Controlling Fall Hazards: Ladders

- Ladder falls are one of the leading causes of workplace fatalities and injuries.
- Three core rules of ladder safety:
 1. Use the right ladder.
 2. Ensure the ladder is free from defects.
 3. Use the ladder properly.



Prep & Inspection

- **Right Tool for the Job:** Always use the correct type, length, and weight rating. Never improvise if a ladder is too short.
- **Mandatory Pre-Check:** Inspect for defects (bent rails, missing parts, grease) before every use.
- **Read the Labels:** Follow all manufacturer instructions and warning stickers.
- **Action:** If a ladder is defective, do not use it. Report it immediately.

Climbing & Working Safely

- **3 Points of Contact:** Always keep two hands and one foot, or two feet and one hand on the ladder.
- **The Belt Buckle Rule:** Keep your weight centered between the side rails. Never lean or overreach—climb down and move the ladder instead.
- **The Danger Zone:** Never stand on the top two steps of a stepladder.



Setup Rules

- **The 4-to-1 Ratio:** When leaning a ladder, place the base 1 foot away from the wall for every 4 feet of working height.
- **The 3-Foot Rule:** When accessing another level (like a roof), the ladder must extend and be secured at least 3 feet above the landing.



Fixed Ladders

- OSHA requires fixed ladders to have a PFAS safety system if the said ladder is installed on or before 12/19/18
- If the ladder is installed before 12/19/18 then it must have a PFAS system, cage and/or a well
- Employers must comply to this before November 18, 2038

Additional Info

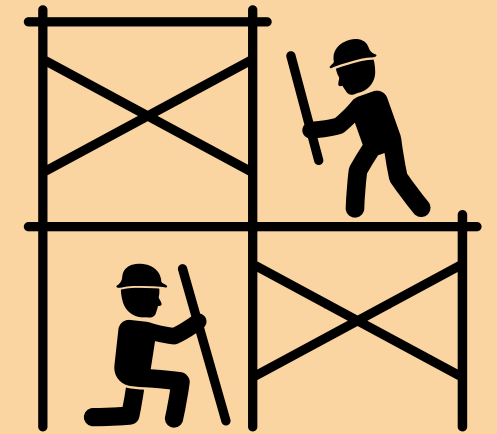
- On April 6, 2026, OSHA proposed to remove the deadline of November 18, 2036. (Docket no. OSHA-2025-0072)
- Under the 2026 guideline employers are allowed to keep well-maintained, pre-existing ladder cages or well until the ladder reaches the end of its service life

Scaffold Setup & Inspection

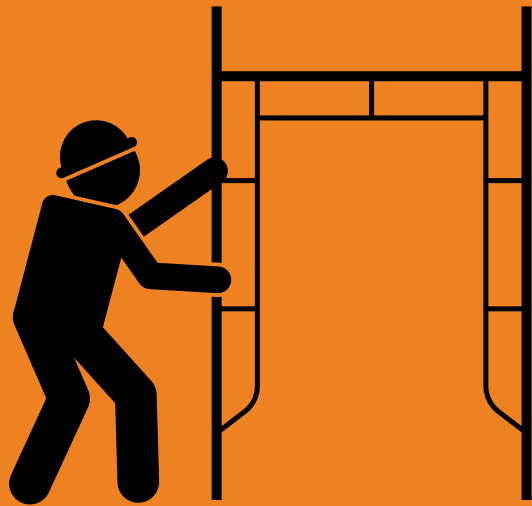
Inspect First: Check for defects before every use. Never use a scaffold with missing or damaged braces, pins, or components.



Mobile Scaffold Setup: The platform height must never exceed 4 times the minimum base dimension (use outriggers if necessary to widen the base).



Full Decking Required: Platforms must be fully planked from front to back. No partial decking or dangerous gaps allowed.



Safe Access & Fall Protection

- **The 10-Foot Rule:** Guardrails or a Personal Fall Arrest System (PFAS) are mandatory on any scaffold 10 feet or higher.
- **Climb Safely:** Only use designated ladders or access points. Never climb the cross-bracing or the scaffold frame.
- **Hands-Free:** Do not carry tools or materials in your hands while climbing on or off the scaffold.

Operating Safely on Scaffolds

- **Keep Your Feet Planted:** Never use a ladder, bucket, or other device on top of a scaffold platform to increase your reach.
- **Powerline Danger:** Keep the scaffold, tools, and materials at least 10 feet away from powerlines (3 feet if insulated).
- **Mobile Scaffold Rules:** Always lock all casters before getting on. Never attempt to move a mobile scaffold while someone is on it.



Scissor Lift Safety & Fall Protection

- Scissor lifts are now officially classified under ANSI standards as Mobile Elevating Work Platforms (MEWPs). Safety compliance now requires specialized MEWP operator training, supervisor training, and documented site risk assessments, rather than just basic scaffold physical controls.
- **The Big Three Hazards:** Most scissor lift injuries are caused by failures in Fall Protection, Stabilization, or Positioning.
- **Guardrails First:** Always verify the guardrail system is fully intact and locked before lifting.
- **Feet on the Floor:** Only stand on the solid work platform. Never stand on or climb the guardrails for extra reach.
- **Don't Overreach:** Position the lift close to your task so you can keep the work within easy

Stair Design & Guarding

Industrial Specs: Fixed stairs must support a 1,000 lb. live load, be at least 22 inches wide, and sit at a 30-50 degree angle.

Uniformity: Tread and riser dimensions must have no more than a $\frac{1}{4}$ inch variation.

The 4-Riser Rule: Handrails (for enclosed stairs) or Stair Rails (for open-sided stairs) are legally required anytime there are 4 or more risers.

Navigating Stairs Sa`ely

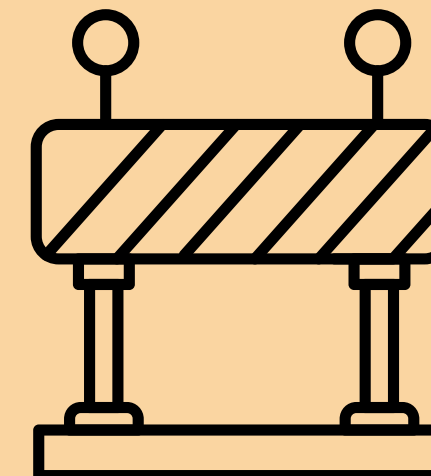
3 Points of Contact: Always keep one hand on the rail (preferably the right side when descending).

Pacing: Never run up or down, and do not jump the last few steps.

Carrying Items: Only carry light loads that do not block your vision or prevent you from holding the handrail.

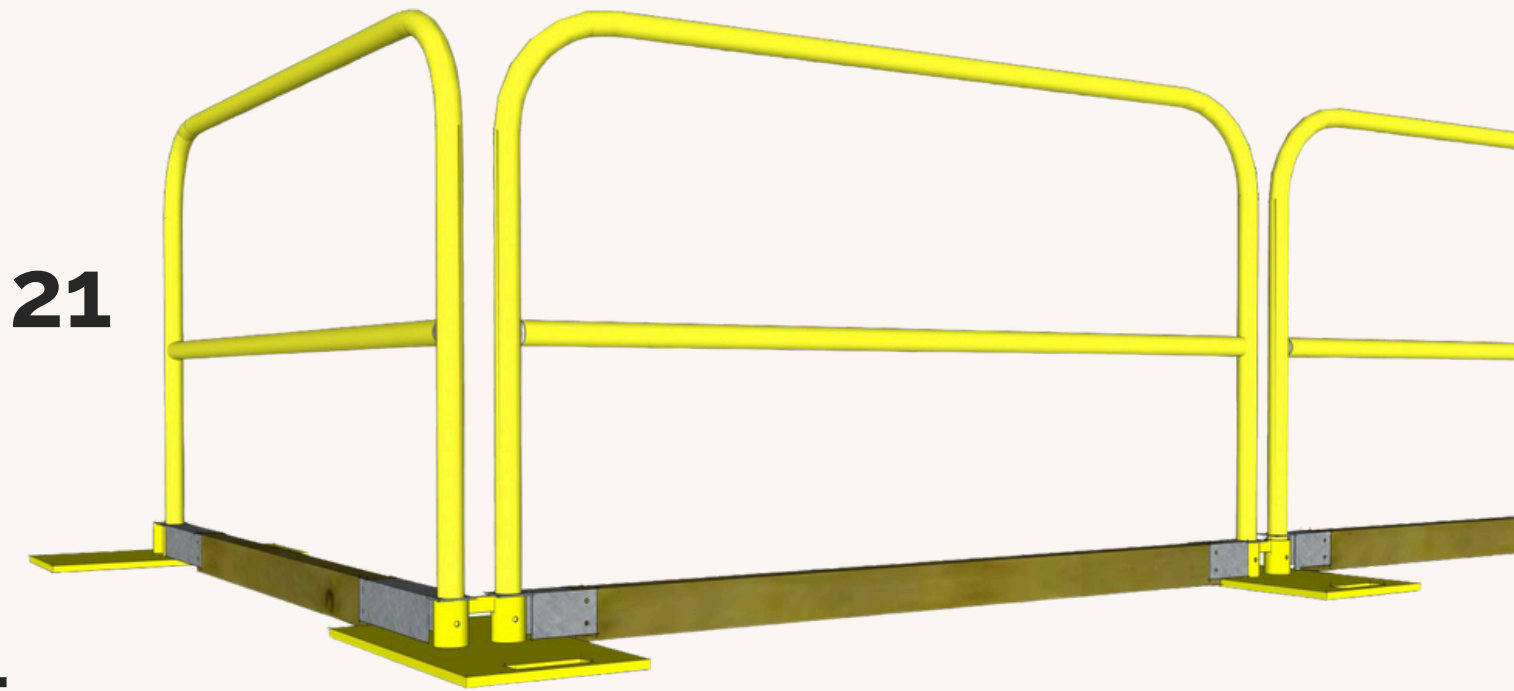
Guarding Floor & Wall Openings

- **Floor Openings:** Must never be left exposed. They require a secure physical cover or a full guardrail system at all times.
- **Temporary Access:** If a floor cover is removed for active work, an attendant (a "guard") must be posted to monitor the opening.
- **Wall Openings & Platforms:** Any wall opening or open-sided platform with a drop of more than 4 feet must be protected by a proper guardrail system.



Guardrail System Specs

- **Top Rail:** Must be 42 inches high (plus or minus 3 inches) from the walking surface.
- **Mid-Rail:** Must be installed halfway up, at 21 inches high.
- **Toeboards (for falling objects):** Must be at least 3.5 inches high, with no more than a $\frac{1}{4}$ -inch gap between the bottom of the board and the floor.



Employer Responsibilities: The Safety Baseline

The Physical Environment

- **The 4-Foot & Zero-Foot Rules:** Employers must install guardrails and toeboards for any open edge 4 feet or higher. However, if working above dangerous machinery, fall protection is required regardless of height (zero feet).
- **Secure the Floor:** Every floor opening must be permanently guarded or securely covered to prevent accidental walk-ins.
- **Housekeeping Mandates:** Work areas must be fundamentally free of known, recognized hazards, and floors must be kept clean, dry, and sanitary.

Worker Support & Compliance

- **Cost-Free PPE:** The employer must select, provide, and maintain all required Personal Protective Equipment (such as safety harnesses, lanyards, and hardhats) at absolutely no cost to the worker.
 - **Comprehensible Training:** Safety training is only legally valid if the worker understands it. Employers are required to deliver training in the specific language and vocabulary their workforce speaks.
- **Alternative Systems:** When standard guardrails are not feasible, employers must provide secondary fall protection like safety nets or a Personal Fall Arrest System (PFAS).